GEOMETRIC REDUCEDNESS AND GEOMETRIC REGULARITY OF THE NORMALIZATION AT A ONE DIMENSIONAL POINT WITH SEPARABLE RESIDUE FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens with the author's own 'We will use the results of Lemma lemma-normalization-locally-algebraic without further mention'; that sentence is reproduced verbatim and the results in question are navigation stubs, while every step that uses them is written out inside the counted span.

Editorial note. The counted spans carry 7 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-normal-in-codim-1 is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Normalization of a locally algebraic scheme: finiteness and dimension behaviour. Same chapter, Lemma lemma-normalization-locally-algebraic, invoked at proof lines 5113-5114. Supplies the finiteness of the normalization and the one dimensionality of the local rings above x.
- A normal one dimensional local ring is a discrete valuation ring. Properties, Lemma properties-lemma-criterion-normal, invoked at proof lines 5116-5118. Supplies the discrete valuation ring structure of the local ring of the normalization.
- Regular local algebras with separable residue field are formally smooth, and formal smoothness equals geometric regularity. More on Algebra, Lemma more-algebra-lemma-regular-implies-fs and Theorem more-algebra-theorem-regular-fs, invoked at proof lines 5119-5126. Supplies geometric regularity of the normalization at the point above x.
- Pointwise criteria for geometric reducedness and geometric regularity. Same chapter, Lemmas lemma-geometrically-reduced-at-point and lemma-geometrically-regular-at-point, invoked at proof lines 5127-5128 and 5135-5136. Supplies the passage between local rings and the geometric properties of the scheme at a point.

Source reference. Original lemma 035W: sources/stacks-varieties/pinned-original-source.tex, source lines 354–354. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 038U: sources/stacks-varieties/pinned-original-source.tex, source lines 2382–2382. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0BXR: sources/stacks-varieties/pinned-original-source.tex, source lines 4903–4903. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 07EJ: sources/stacks-chow/tags.txt, source lines 9449–9449. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original theorem 07EL: sources/stacks-chow/tags.txt, source lines 9451–9451. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0345: sources/stacks-chow/tags.txt, source lines 4080–4080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 06LF: sources/stacks-varieties/pinned-original-source.tex, source lines 3713–3713. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme. Let $\nu : X^\nu \to X$ be the normalization of X. Let $x \in X$ be a point such that (a) $\mathcal{O}_{X,x}$ is reduced, (b) $\dim(\mathcal{O}_{X,x}) = 1$, and (c) for every $x' \in X^\nu$ with $\nu(x') = x$ the extension $\kappa(x')/k$ is separable. Then X is geometrically reduced at x and X^ν is geometrically regular at x' with $\nu(x') = x$.*

Proof. We will use the results of Lemma 0BXR without further mention. Let $x' \in X^\nu$ be a point over x. By dimension theory (Section 06LF) we have $\dim(\mathcal{O}_{X^\nu,x'}) = 1$. Since X^ν is normal, we see that $\mathcal{O}_{X^\nu,x'}$ is a discrete valuation ring (Properties, Lemma 0345). Thus $\mathcal{O}_{X^\nu,x'}$ is a regular local k-algebra whose residue field is separable over k. Hence $k \to \mathcal{O}_{X^\nu,x'}$ is formally smooth in the $\mathfrak{m}_{x'}$-adic topology, see More on Algebra, Lemma 07EJ. Then $\mathcal{O}_{X^\nu,x'}$ is geometrically regular over k by More on Algebra, Theorem 07EL. Thus X^ν is geometrically regular at x' by Lemma 038U.

Since $\mathcal{O}_{X,x}$ is reduced, the family of maps $\mathcal{O}_{X,x} \to \mathcal{O}_{X^\nu,x'}$ is injective. Since $\mathcal{O}_{X^\nu,x'}$ is a geometrically reduced k-algebra, it follows immediately that $\mathcal{O}_{X,x}$ is a geometrically reduced k-algebra. Hence X is geometrically reduced at x by Lemma 035W. □